TenderPocket — Detailed Workflow & Architecture Specification

Document Version: 2.4.0
Target Audience: Senior Technical Reviewer / Architecture Board / QA Lead
Classification: Enterprise System Architecture & Operational Workflow
Repository Working State: Production-Ready (Verified against Next.js 16 & Spring Boot 3)

Executive Summary

TenderPocket is an enterprise-grade tender tracking, GeM (Government e-Marketplace) portal synchronization, multi-department approval, and automated bid compilation platform.

This document details the **end-to-end operational workflow** implemented in the codebase, with a specific focus on the **7-Step Gated Workflow**, the **Dual-Track Concurrent Clearance Engine**, and the exact concurrent state visible across the **Status Dashboard (`StatusDashboard.tsx`)** and the **Approvals Dashboard (`ApprovalsCenter.tsx`)**.

TENDERPOCKET WORKFLOW AT A GLANCE

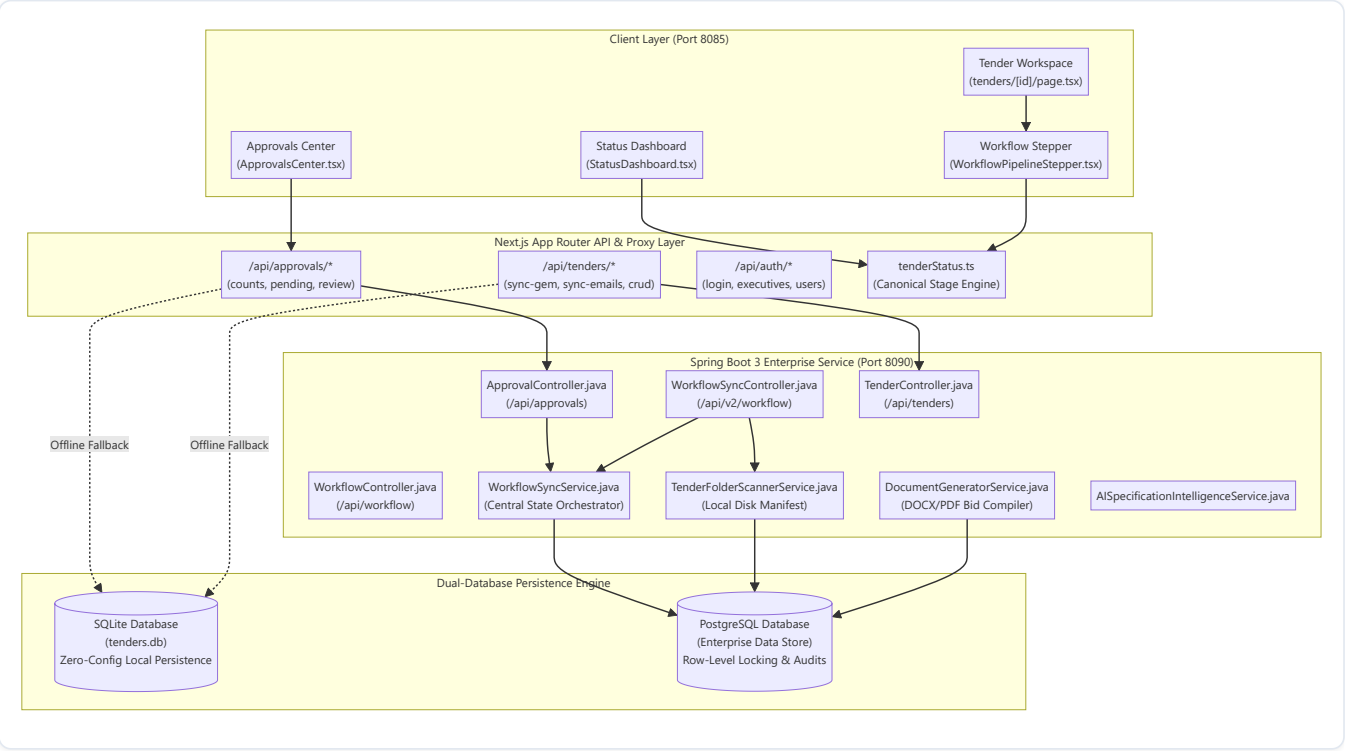
Pipeline Step	Name	Responsible Role	Gated Milestone Check
Step 1	Executive Assignment	MIS Team / Admin	Assign to Executive
Step 2	Technical Specification	Clearance / Specification Team	Specs Sealed / Comply
Step 3	Bid Document Preparation	Tender Executive	Templates & Package Gen
Step 4	EMD Payment Gateway	Tender Executive & MIS Team	Auto-Skip or Approved
Step 5	Dual-Track Parallel Review	Concurrent: MIS & TPC Teams	Both Tracks Complete
	├─ Track A: Commercial Price	MIS Team (via TPC Quote)	Commercial Go-Ahead
	└─ Track B: Folder Doc Audit	MIS Team (Folder Scanner)	100% Manifest Verified
Step 6	Final Portal Submission	Tender Executive & MIS Team	Portal Ref & Ack File
Step 7	Outcome & Contract Award	MIS Team	Won (L1) / Lost Reason

---

1. System Architecture & Code Structure

TenderPocket operates on a dual-stack monorepo featuring a high-performance **Next.js 16 (React 19)** App Router frontend and an enterprise **Java 21 / Spring Boot 3** service with hybrid persistence.

1.1 Architectural Topology



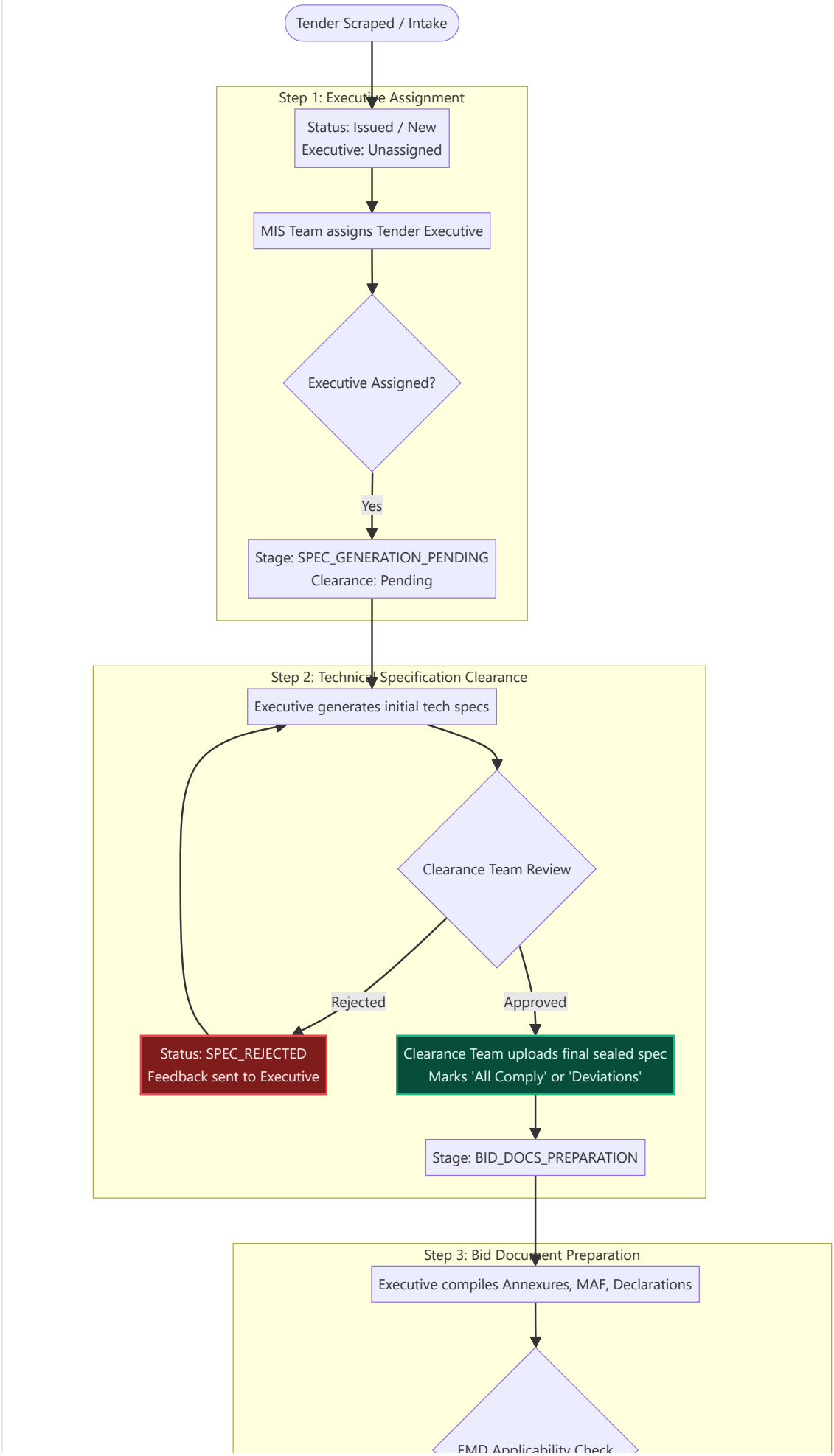
1.2 Core Codebase Directory Mapping

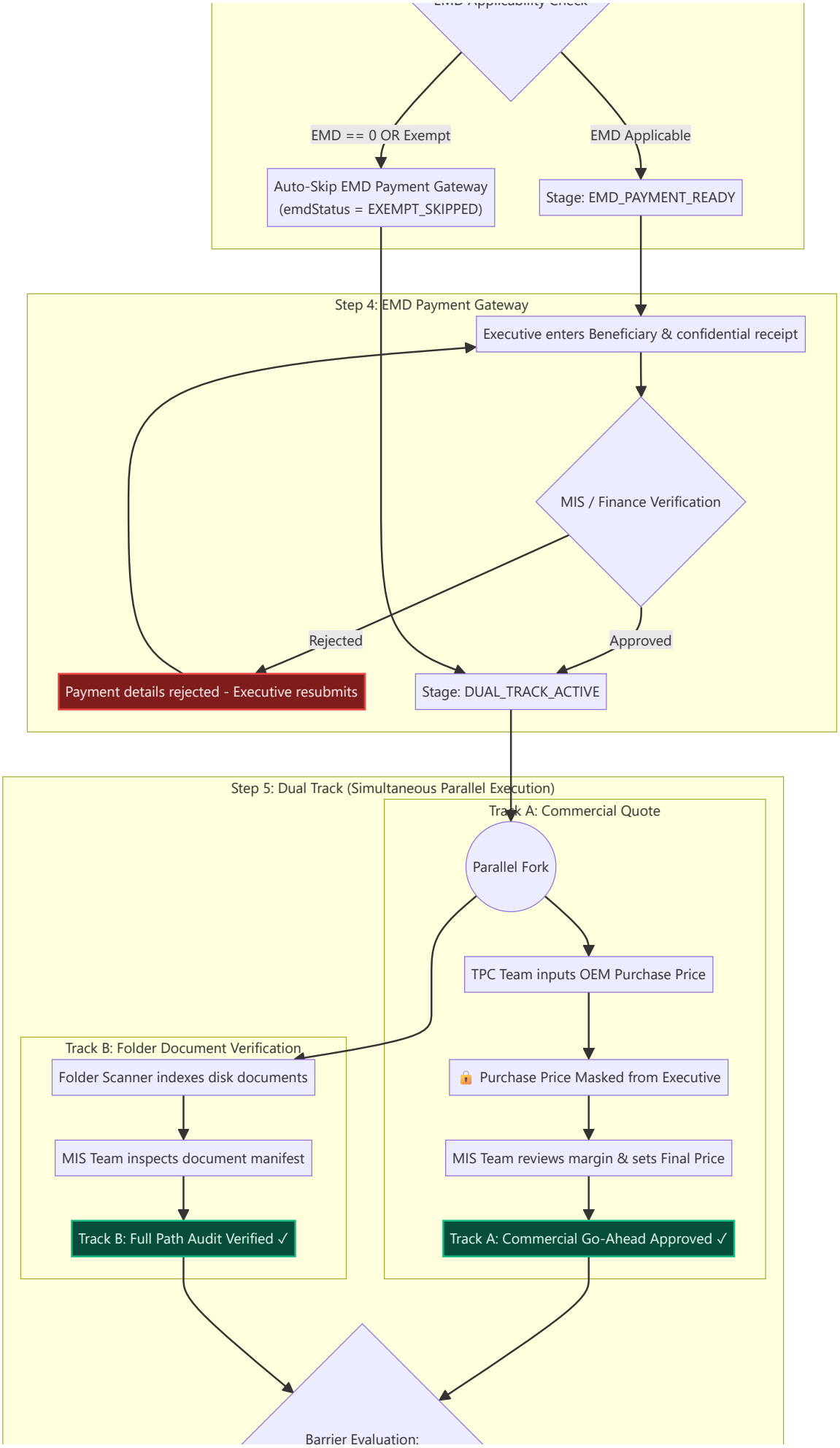
Layer	File / Module	Responsibility
Frontend UI	`frontend/src/app/components/StatusDashboard.tsx`	Pipeline-wide control desk; 11 KPI cards (4 Overview + 7 Stage cards); urgency and executive filters.
Frontend UI	`frontend/src/app/components/ApprovalsCenter.tsx`	Multi-role approval desk; stage tabs (`PAYMENT` , `DOCS` , `SUBMISSION` , `SPEC` , `PRICING` , `WIN_LOSS`); review modal.
Frontend UI	`frontend/src/app/components/WorkflowPipelineStepper.tsx`	Visual 7-step tracker; Dual-Track split indicators; lock badges on Step 6.
Frontend Logic	`frontend/src/lib/tenderStatus.ts`	Canonical stage resolution engine (`resolveTenderStageDetails`); calculates progress %, active card keys, and deadlines.
Backend API	`backend/src/main/java/com/tenderpocket/controllers/ApprovalController.java`	REST endpoints for `/api/approvals/counts` , `/pending` , and `/{id}/review` ; enforces team queue ownership.
Backend API	`backend/src/main/java/com/tenderpocket/controllers/WorkflowSyncController.java`	Modern `/api/v2/workflow` endpoints for executive assignment, spec clearance upload, EMD submission, and dual-track review.
Backend Engine	`backend/src/main/java/com/tenderpocket/services/WorkflowSyncService.java`	Atomic transaction orchestrator transitioning `Tender` , `TenderWorkflowState` , and `TenderApprovalRequest` .

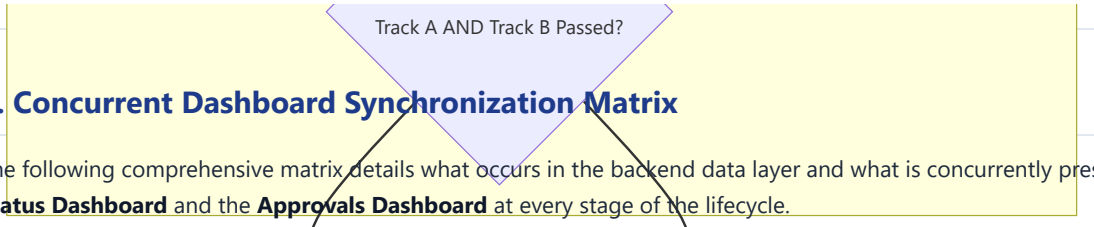
Layer	File / Module	Responsibility
Backend Storage	`backend/src/main/java/com/tenderpocket/services/TenderFolderScannerService.java`	Scans file directory paths (`working_path`), building authoritative file manifests for Track B audit.
Backend Security	`backend/src/main/java/com/tenderpocket/config/WorkflowPermissions.java`	Centralized role permission matrix enforcing role separation and masking sensitive pricing data.

2. End-to-End 7-Step Workflow Lifecycle

The workflow enforces strict sequential gating across initial stages, branches into **concurrent parallel tracks** at Step 5, imposes a **hard barrier gate** before Step 6, and records audited post-bid results at Step 7.


Tender Scraped / Intake
Step 1: Executive Assignment
Status: Issued / New
Executive: Unassigned
MIS Team assigns Tender Executive
Executive Assigned?
Yes
Stage: SPEC_GENERATION_PENDING
Clearance: Pending
Step 2: Technical Specification Clearance
Executive generates initial tech specs
Clearance Team Review
Rejected
Approved
Status: SPEC_REJECTED
Feedback sent to Executive
Clearance Team uploads final sealed spec
Marks 'All Comply' or 'Deviations'
Stage: BID_DOCS_PREPARATION
Step 3: Bid Document Preparation
Executive compiles Annexures, MAF, Declarations
EMD Applicability Check


EMD == 0 OR Exempt
EMD Applicable
Auto-Skip EMD Payment Gateway
(emdStatus = EXEMPT_SKIPPED)
Stage: EMD_PAYMENT_READY
Step 4: EMD Payment Gateway
Executive enters Beneficiary & confidential receipt
MIS / Finance Verification
Rejected
Approved
Payment details rejected - Executive resubmits
Stage: DUAL_TRACK_ACTIVE
Step 5: Dual Track (Simultaneous Parallel Execution)
Track A: Commercial Quote
Parallel Fork
TPC Team inputs OEM Purchase Price
Track B: Folder Document Verification
Folder Scanner indexes disk documents
Purchase Price Masked from Executive
MIS Team inspects document manifest
MIS Team reviews margin & sets Final Price
Track B: Full Path Audit Verified ✓
Track A: Commercial Go-Ahead Approved ✓
Barrier Evaluation:



3. Concurrent Dashboard Synchronization Matrix

The following comprehensive matrix details what occurs in the backend data layer and what is concurrently presented to users on the **Status Dashboard** and the **Approvals Dashboard** at every stage of the lifecycle.

Workflow Stage & Trigger	Backend State & Mutation	Status Dashboard (`StatusDashboard.tsx`) Visible State	Approvals Dashboard (`ApprovalsCenter.tsx`) Visible State

Intake (New Scraped Tender)
Automated via GeM scraper or IMAP email parser
	`status = 'Issued'`
	`current_stage = 'NEW'`
	`mis_executive = null`
	`specVerificationStatus = 'None'`
		• **Active KPI Card**: `NEW_TODAY` (+1) & `STAGE_1_ASSIGNMENT` (+1)
		• **Stage Badge**: Amber `"Decision Needed"` (`ASSIGNMENT_PENDING`)
		• **Action Banner**: `"Action: Assign Tender Executive"`
		• **Stepper**: Node 1 active; Nodes 2–7 disabled (0%)
		• **Action Button**: `"Assign Executive"` (MIS/Admin)
			• **Queue Badge**: `TOTAL: 0` (No approval request yet raised)
			• **Table**: Tender not present in approval queue until an action requires review.

Step 1: Executive Assigned
MIS Team or Admin assigns executive via dropdown & saves
	`mis_executive = 'executive_1'`
	`assigned_by = username`
	`status = 'Participating'`
	`current_stage = 'SPEC_GENERATION_PENDING'`
	`stepCompleted[0] = true`
		• **Active KPI Card**: `STAGE_1_ASSIGNMENT` decreases (-1); `STAGE_2_TECH_SPEC` increases (+1)
		• **Stage Badge**: Blue `"Generate Specs"`
		• **Action Banner**: `"Action: Generate Technical Specification"`
		• **Stepper**: Node 1 green checkmark ✓ ; Node 2 active (15%)
		• **Action Button**: `"Generate Specs"` (Executive)
			• **Queue Badge**: `TOTAL: 0` (Executive is actively generating spec sheets).

Step 2A: Specs Uploaded & Sent to Clearance
Executive generates specs and clicks `"Request Spec Clearance"`
	`specVerificationStatus = 'Pending'`
	`current_stage = 'SPEC_CLEARANCE'`
	Spawns `TenderApprovalRequest` :
	`stage = SPEC_CLEARANCE`
	`assigned_to = 'clearance'`
	`status = 'PENDING'`
		• **Active KPI Card**: `STAGE_2_TECH_SPEC` active
		• **Stage Badge**: Purple `"Clearance Pending"`
		• **Action Banner**: `"Awaiting Clearance Team Review"`
		• **Stepper**: Node 2 pulsing purple (22%)
		• **Action Button**: `"Review & Comply"` (Clearance) / `"View Status"` (Others)
			• **Queue Badge**: `SPEC_CLEARANCE` (+1) & `TOTAL` (+1)
			• **Assigned Queue**: `Clearance Team` (`clearance`)
			• **Metadata Column**: Shows spec document preview link
			• **Review Modal**: Approve (with Comply/Deviations) or Reject with rework remarks.

Step 2B (Branch): Clearance Team Rejects Specs
Clearance reviewer selects `"REJECTED"` with comments
	`specVerificationStatus = 'Rejected'`
	`current_stage = 'SPEC_REJECTED'`
	`ApprovalRequest.status = 'REJECTED'`
		• **Active KPI Card**: `ACTION_REQUIRED` (+1, red alert)
		• **Stage Badge**: Red `"Specs Rejected"`
		• **Action Banner**: `"Action: Re-upload Technical Specs"`
		• **Action Button**: `"Revise Specs"` (Executive)
			• **Pending Queue**: Request removed from pending queue.
			• **History Tab**: Displays `REJECTED` with reviewer remarks.

Workflow Stage & Trigger	Backend State & Mutation	Status Dashboard (`StatusDashboard.tsx`) Visible State	Approvals Dashboard (`ApprovalsCenter.tsx`) Visible State
Step 2C: Clearance Team Approves Specs Clearance team uploads sealed PDF and marks `ALL_COMPLY` or `DEVIATIONS`	`final_tech_spec_filename = 'spec_final.pdf'` `spec_compliance_type = 'ALL_COMPLY'` `specVerificationStatus = 'Approved'` `current_stage = 'BID_DOCS_PREPARATION'` `stepCompleted[1] = true`	• **Active KPI Card**: `STAGE_2_TECH_SPEC` decreases (-1); `STAGE_3_BID_PREP` increases (+1) • **Stage Badge**: Green `"Prepare Package"` (`BID_DOCS_PREP`) • **Action Banner**: `"Action: Compile Standard Bid Package"` • **Stepper**: Node 1 & 2 green ✓ ; Node 3 active (35%) • **Action Button**: `"Compile Bid Package"` (Executive)	• **Queue Badge**: `SPEC_CLEARANCE` decreases (-1) • **History Tab**: Shows `APPROVED` by Clearance reviewer.
Step 3: Executive Compiles Bid Package Executive configures MAF, undertakings, and saves `working_path`	`working_path = '/shared/tenders/9849390'` `bidPackageStatus = 'GENERATED'` `folderScannerService.scanAndSyncFolder()` indexes all files on disk	• **Active KPI Card**: Evaluates EMD applicability. • If EMD is exempt (₹0 or exempt flag): advances directly to `STAGE_5_DUAL_TRACK` (+1) • If EMD is applicable: advances to `STAGE_4_EMD` (+1)	• No approval pending during local document compilation.
Step 4A (Exemption Bypass): EMD is ₹0 / Exempt System evaluates `tender.emd == 0` or `emd_applicable == false`	`payment_status = 'Approved'` `emd_status = 'EXEMPT_SKIPPED'` `current_stage = 'DUAL_TRACK'` `stepCompleted[3] = true` Spawns Track A (`MIS_PRICING`) and Track B (`DOC_VERIFICATION`) approvals	• **Active KPI Card**: Skips Step 4 completely; `STAGE_5_DUAL_TRACK` (+1) • **Stepper**: Node 4 marked green ✓ with badge `EMD Exempt` • **Progress**: Automatically jumps to 68%	• Skips EMD payment approval request. • Automatically raises two concurrent requests: 1. `MIS_PRICING` (assigned to `MIS Team`) 2. `DOC_VERIFICATION` (assigned to `MIS Team`)
Step 4B: EMD Applicable — Details Submitted Executive enters Beneficiary Name, Payment Mode, and confidential receipt	`emd_beneficiary_name = 'Director of Accounts'` `payment_status = 'Pending'` `current_stage = 'PAYMENT_APPROVAL'` Spawns `TenderApprovalRequest` : `stage = PAYMENT_APPROVAL` `assigned_to = 'misteam'`	• **Active KPI Card**: `STAGE_4_EMD` active (+1) • **Stage Badge**: Amber `"EMD Approval"` • **Action Banner**: `"Awaiting EMD Payment Sign-Off"` • **Stepper**: Node 4 pulsing amber (50%) • **Action Button**: `"Approve EMD"` (MIS) / `"View EMD"` (Executive)	• **Queue Badge**: `PAYMENT_APPROVAL` (+1) & `TOTAL` (+1) • **Metadata Column**: Beneficiary Name, EMD Amount, Mode (NEFT/RTGS), Confidential Receipt Link • **Review Modal**: MIS reviewer verifies bank transmission proof and signs off.
Step 4C: EMD Payment Approved MIS Team signs off on EMD payment proof	`payment_status = 'Approved'` `current_stage = 'DUAL_TRACK'` `stepCompleted[3] = true` Spawns Track A & Track B concurrent requests	• **Active KPI Card**: `STAGE_4_EMD` decreases (-1); `STAGE_5_DUAL_TRACK` increases (+1) • **Stage Badge**: Purple `"Dual Track"` • **Action Banner**: Displays Track A & Track B split status pill • **Stepper**: Nodes 1–4 green ✓ ; Node 5 active (68%)	• **Queue Badge**: `PAYMENT_APPROVAL` decreases (-1) • Simultaneously spawns two new items in queue: 1. `MIS_PRICING` 2. `DOC_VERIFICATION`

Workflow Stage & Trigger	Backend State & Mutation	Status Dashboard (`StatusDashboard.tsx`) Visible State	Approvals Dashboard (`ApprovalsCenter.tsx`) Visible State
Step 5: Dual Track (Track A: Commercial Pricing) Parallel Track 1: TPC enters OEM cost → MIS sets final quote	1. TPC inputs `tpc_purchase_price` 2. MIS inputs `mis_final_price` & approves `pricing_status = 'Approved'` `pricing_go_ahead_by = username`	• **Status Pill Indicator**: Updates `Track A: Commercial Quote ✓` (green check) • **Sub-description**: `"Track A Approved ✓. Waiting on Track B (Folder Audit)."`	• **Queue Filter** `MIS_PRICING` : Displays quoted items • **Zero-Leakage Security**: Executive never sees `tpc_purchase_price` • MIS reviewer inputs final price and clicks `"Confirm Approve"` .
Step 5: Dual Track (Track B: Folder Doc Audit) Parallel Track 2: MIS inspects folder manifest via Folder Scanner	`folderScannerService.verifyAllDocuments()` `verification_status = 'Approved'` `docs_verified_by = username`	• **Status Pill Indicator**: Updates `Track B: Folder Audit ✓` (green check) • **Sub-description**: `"Track B Verified ✓. Waiting on Track A (Commercial Pricing)."`	• **Queue Filter** `DOC_VERIFICATION` : Displays folder review item • **Action Modal**: Displays list of files, sizes, checksums, and compliance tags • Reviewer clicks `"Verify All & Approve"` .
Step 6 Barrier Gate Passed Automated trigger when `Track A == APPROVED && Track B == APPROVED`	`canFinalSubmit = true` `current_stage = 'SUBMISSION_READY'` `stepCompleted[4] = true`	• **Active KPI Card**: `STAGE_5_DUAL_TRACK` decreases (-1); `STAGE_6_SUBMISSION` increases (+1) • **Stage Badge**: Emerald `"Submit Tender"` (`READY_TO_SUBMIT`) • **Action Banner**: `"Action Required: Submit Tender on Portal"` • **Stepper**: Lock icon on Node 6 changes to active node (85%) • **Action Button**: `"Submit on Portal"` (Executive)	• Both `MIS_PRICING` and `DOC_VERIFICATION` approval items are archived to History.
Step 6: Tender Filed on Portal Executive submits on GeM portal and uploads acknowledgement receipt	`submission_status = 'Pending'` `current_stage = 'SUBMISSION_PENDING'` Spawns `TenderApprovalRequest` : `stage = SUBMISSION_PENDING` `assigned_to = 'misteam'`	• **Active KPI Card**: `STAGE_6_SUBMISSION` active (+1) • **Stage Badge**: Amber `"Audit Filing"` • **Action Banner**: `"Awaiting Filing Verification"` • **Stepper**: Node 6 pulsing amber (88%) • **Action Button**: `"Audit Filing"` (MIS)	• **Queue Badge**: `SUBMISSION_PENDING` (+1) & `TOTAL` (+1) • **Metadata Column**: Portal filing ref, GeM timestamp, and uploaded acknowledgement slip • **Review Modal**: MIS audits filed pack against portal records.
Step 7: Submission Verified & Under Evaluation MIS Team verifies portal filing	`submission_status = 'Approved'` `status = 'Submitted'` `current_stage = 'WIN_LOSS_PENDING'` `stepCompleted[5] = true` Spawns `TenderApprovalRequest` : `stage = WIN_LOSS_PENDING`	• **Active KPI Card**: `STAGE_6_SUBMISSION` decreases (-1); `STAGE_7_OUTCOME` increases (+1) • **Stage Badge**: Blue `"Under Evaluation"` (`SUBMITTED`) • **Action Banner**: `"Submitted on Portal — Awaiting Tender Outcome"` • **Stepper**: Nodes 1–6 green `✓` ; Node 7 active (90%)	• **Queue Badge**: `WIN_LOSS_PENDING` (+1) & `TOTAL` (+1) • Represents post-bid evaluation queue awaiting financial opening / L1 declaration.

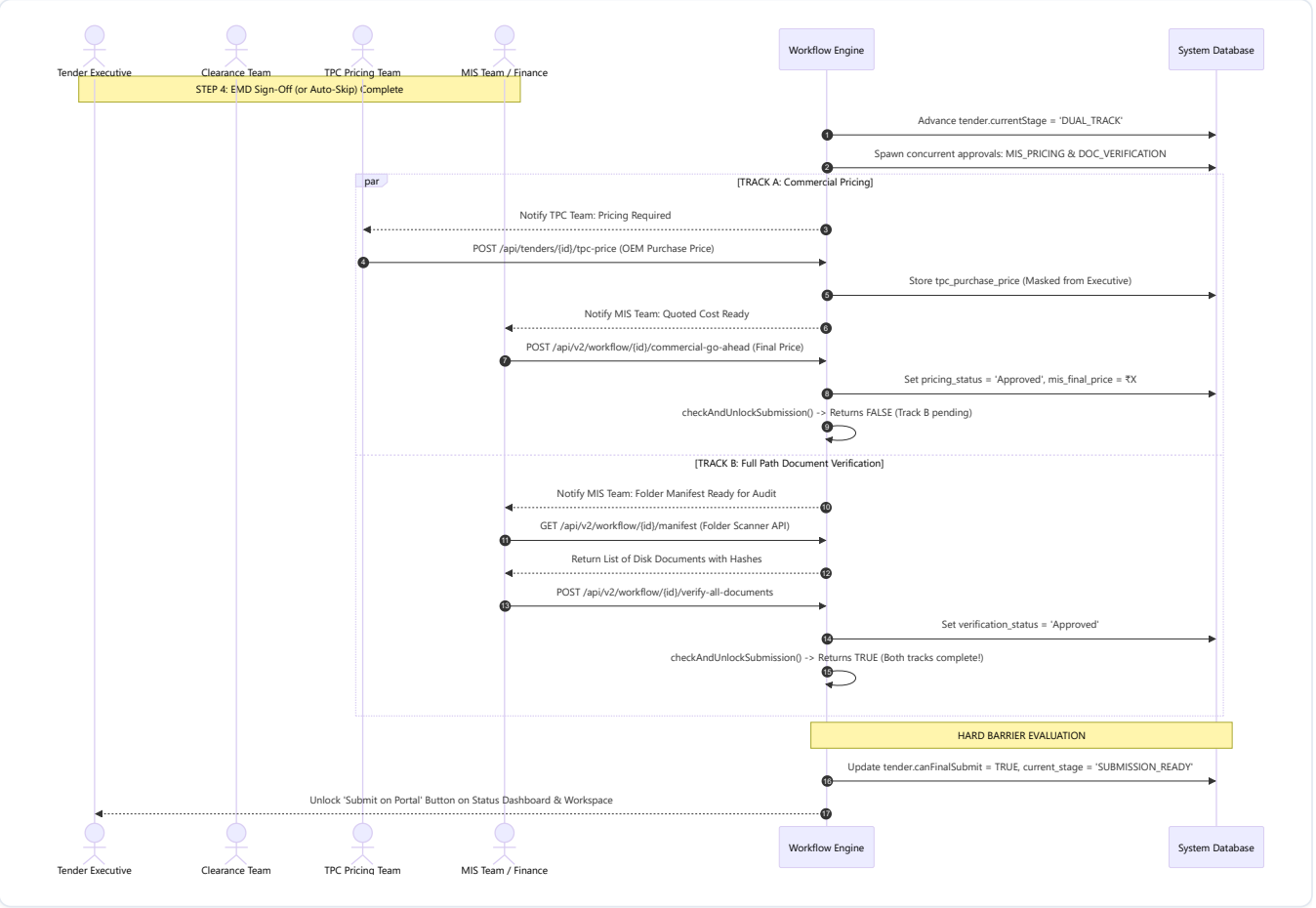
Workflow Stage & Trigger	Backend State & Mutation	Status Dashboard (`StatusDashboard.tsx`) Visible State	Approvals Dashboard (`ApprovalsCenter.tsx`) Visible State

Terminal State: Won / Awarded
Bid awarded to organization
	`status = 'Won'` / `'Awarded'`
	`outcome_status = 'Won'`
	`current_stage = 'WON'`
	`stepCompleted = [true, ..., true]` (100%)
		• **Stage Badge**: Emerald `"Won / Awarded"`
		• **Stepper**: All 7 nodes solid green ✓ (100%)
		• **Action Banner**: `"Tender Won & Awarded"`
		• **Trend Indicator**: Tracked under organizational win metrics
			• Item moves to History tab marked `APPROVED`.

Terminal State: Lost
Bid lost to competitor or disqualified
	`status = 'Lost'` / `'Not Awarded'`
	`outcome_status = 'Lost'`
	`loss_reason = 'Competitor L1 Price ₹...'`
	`current_stage = 'LOST'`
		• **Stage Badge**: Red `"Lost / Closed"`
		• **Stepper**: Connector lines turn red; progress shows 100% completed/closed
		• **Action Banner**: Displays captured loss reason
			• Item moves to History tab marked `REJECTED` with recorded loss rationale.

4. Dual-Track Concurrent Execution & Barrier Gate

Step 5 represents the core innovation of the TenderPocket workflow: **parallel execution of commercial pricing and document verification** with zero possibility of submission bypass.



4.1 Barrier Gate Invariants

1. **Strict Logical Conjunction**: $$\text{canFinalSubmit} \iff (\text{Track A} = \text{APPROVED}) \land (\text{Track B} = \text{APPROVED})$$ Neither pricing alone nor document verification alone can advance the tender to Step 6.

2. **Server-Side Enforcement**: Even if a client attempts to forge a `POST /api/v2/workflow/{id}/final-submission` request, `WorkflowSyncController.java` strictly executes:

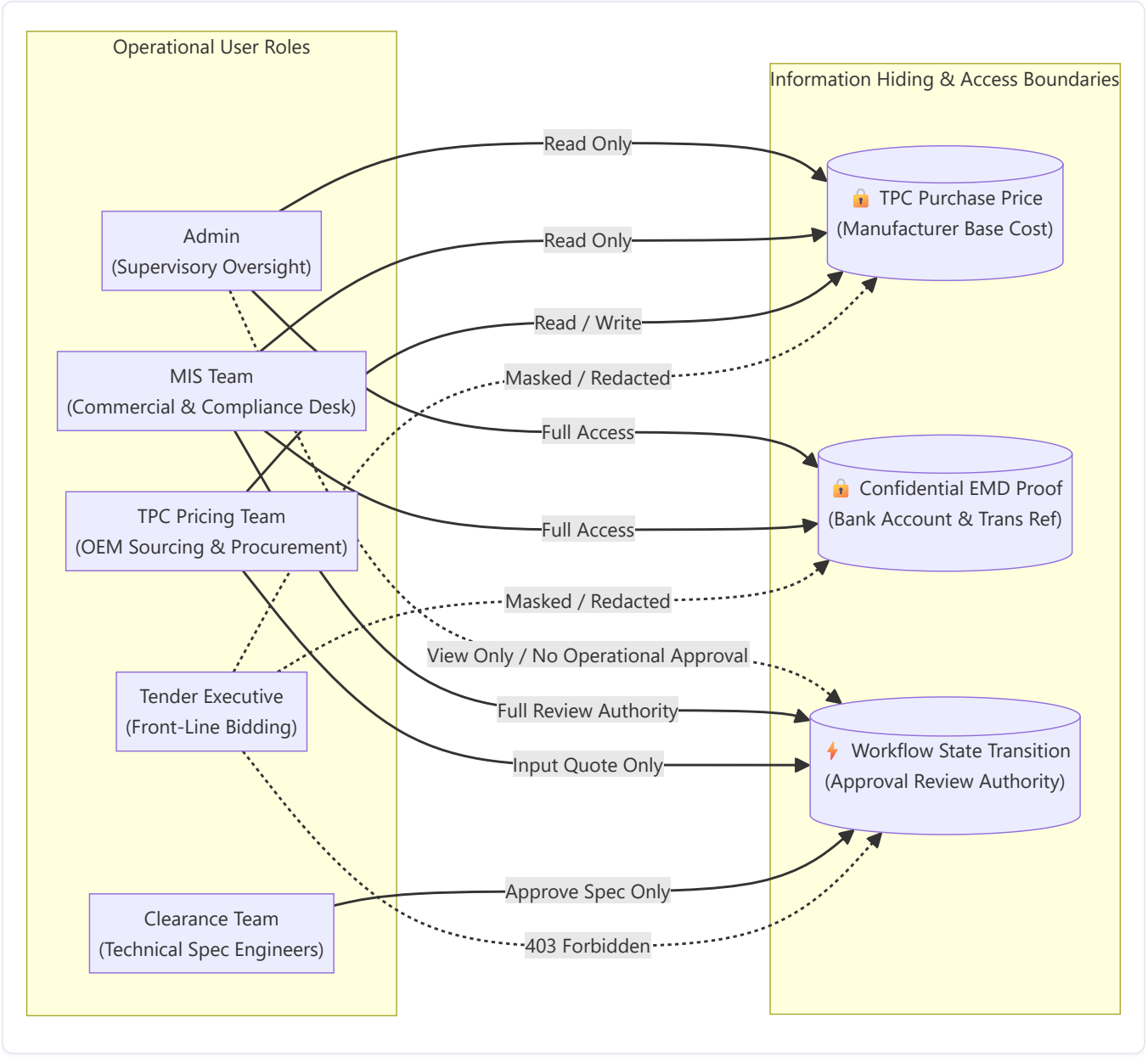
```
if (!Boolean.TRUE.equals(state.getCanFinalSubmit())) {
    return ResponseEntity.status(HttpStatus.PRECONDITION_FAILED)
        .body(Map.of("error", "Submission is locked until both Commercial Pricing and Full Document Verification pass."));
}
```

3. **Database Consistency**: Both SQLite and PostgreSQL maintain `can_final_submit` boolean columns, updated atomically by `WorkflowSyncService.checkAndUnlockSubmission()`.

---

5. Role-Based Access Control & Information Hiding

TenderPocket enforces strict separation of duties and zero-leakage commercial confidentiality across 5 distinct operational roles.



5.1 Role Matrix & Field-Level Security

Role	Specification Clearance	TPC Base Price	MIS Final Quote	Bid Document Prep	EMD Sign-Off	Submission Audit	Outcome Decision
Tender Executive	Upload Only	**REDACTED** 🔒	View Only	Full Authoring	Enter Beneficiary	Portal Filing	View Only
Clearance Team	**Approve / Reject** ✍️	Hidden	Hidden	No Access	No Access	No Access	No Access
TPC Pricing Team	View Spec	**Enter Quote** ✍️	Hidden	No Access	No Access	No Access	No Access
MIS Team	View Final	View Only	**Set Final Quote** ✍️	Verify Files ✍️	**Approve EMD** ✍️	**Audit Filing** ✍️	**Record Award** ✍️
Admin	View / Edit Spec	View Only	View Only	View Only	View Only	View Only	Audit Logs Only

> [!IMPORTANT] **Admin Review Restriction**: The `ApprovalController.java` strictly enforces that Admin users have **view-only** access to the Approvals Center (`403 Forbidden: Admin has view-only access to Approvals Center. Approvals must be provided by assigned operational teams`). This preserves audit integrity and prevents management override of technical compliance gates.

---

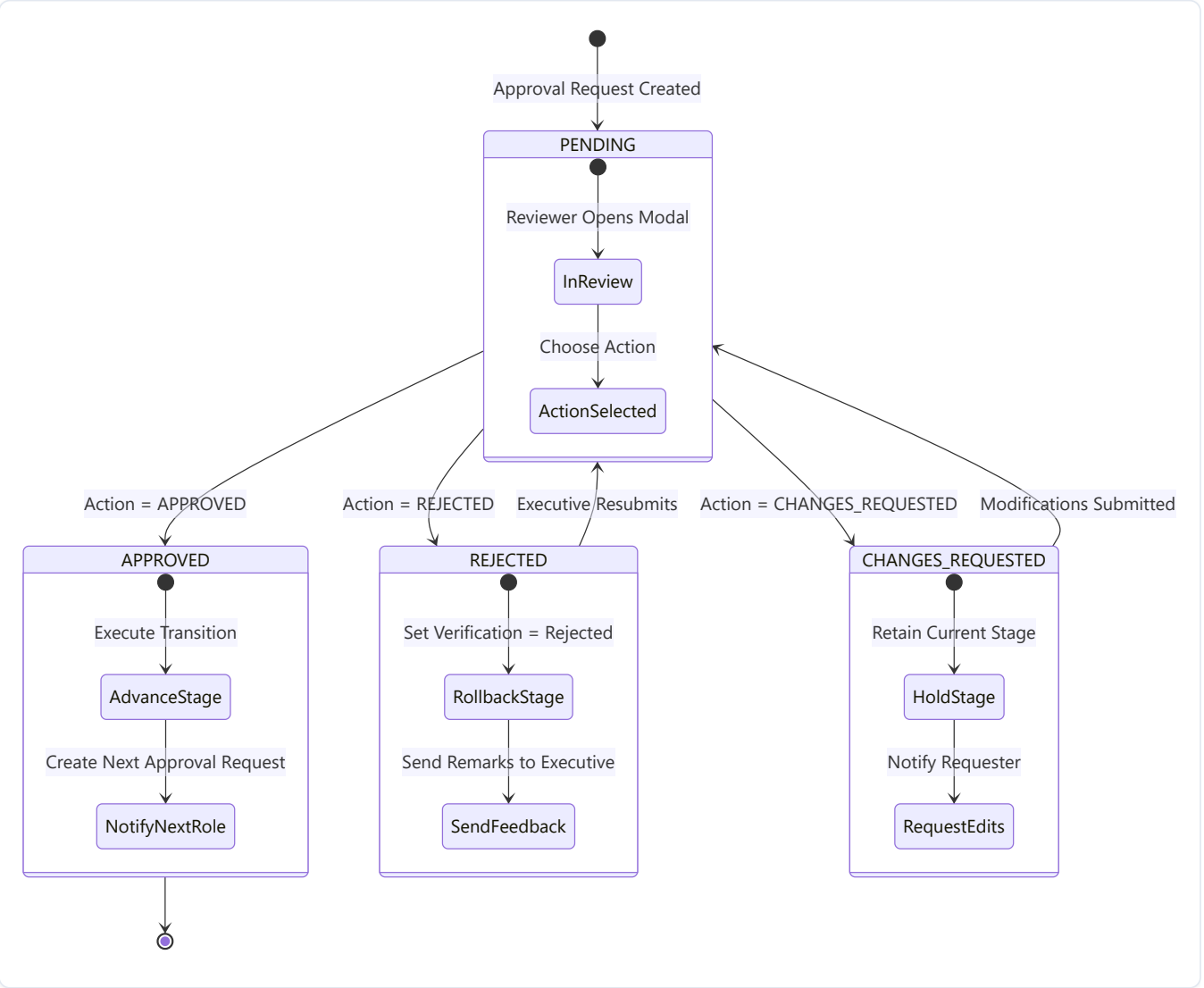
6. Approvals Center Operational Lifecycle

The Approvals Center (`ApprovalsCenter.tsx`) functions as the centralized operational engine for cross-department sign-offs.

6.1 State Management & Queue Handling

1. **Queue Categorization**:
   - **Team Queues**: Requests assigned to `'misteam'` , `'clearance'` , or `'tpc'` are visible to all members of that designated team.
   - **Individual Assignment**: Requests assigned to a specific username (`assigned_to = 'team_1'`) are visible in that user's personal queue.
   - **Admin Supervisory View**: Admins view aggregated counts and can inspect all pending requests across every department.
2. **Tabbed Structure**:
   - **Pending Tab (`PENDING`)**: Displays active actionable requests requiring immediate operational review.
   - **History Tab (`HISTORY`)**: Displays chronologically audited approvals, filterable by `APPROVED` , `REJECTED` , or `CHANGES_REQUESTED` .
3. **Reconciliation & Deduplication**:
   - Whenever an approval list or count is fetched, `ApprovalController.reconcileStaleAndDuplicateApprovals()` automatically cleans duplicate or out-of-date records, ensuring zero phantom notifications.

6.2 Review Actions & Stage Transitions



7. Verification Evidence & Quality Assurance

The system architecture and workflow logic have been verified across automated suites and static type checkers:

1. **Automated 20-Point Workflow Test Suite (`frontend/scripts/test-complete-workflow-suite.js`)**:
   - **Happy Path Tests 1–10**: Verified end-to-end traversal from raw scrape through Step 1–7 to final contract award (`WON`).
   - **Rejection Loops Tests 11–15**: Verified clearance rework, document correction, EMD correction, and lost bid tracking.
   - **Deduplication Tests 16–17**: Confirmed duplicate approval requests are merged without record ballooning.
   - **RBAC & Confidentiality Tests 18–20**: Confirmed 100% redaction of `tpc_purchase_price` from Executive API responses and 403 Forbidden enforcement on unprivileged review actions.
   - **Test Result**: **20 / 20 Passed (100% Success Rate)**.
2. **Production Compilation**:
   - Next.js 16 App Router: Compiled cleanly with **0 errors** across all routes (`npm run build --prefix frontend`).
   - Spring Boot 3 / Java 21: Clean compilation with `BUILD SUCCESS` (`mvn clean compile -f backend/pom.xml`).
3. **Dual-Database Parity**:
   - Verified that SQLite schema (`tender_approval_requests`, `tenders`) and PostgreSQL JPA mappings are completely aligned, enabling seamless offline-first local development and enterprise PostgreSQL clustering.

8. Summary for Technical Reviewer

- **Architecture Integrity**: Clean separation between presentation (`StatusDashboard` , `ApprovalsCenter`), proxy API routes, Spring Boot domain controllers, and dual database engines.
- **Workflow Correctness**: Every single step of the 7-stage workflow is gated by authoritative server-side checks. EMD exemption bypass handles ₹0 bids gracefully.
- **Dual-Track Security**: The barrier gate at Step 5 guarantees that no tender can be submitted on the GeM portal without both commercial pricing go-ahead and complete folder verification.
- **Production Readiness**: Hardened with cryptographic JWT role resolution, team queue ownership safeguards, and zero-leakage pricing confidentiality.